MATH441/MATH541
Applied Mathematics II: Complex Variables
Lecture slides 22: Examples of Real Integrals

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Lecture outline

1. Examples

Example 1. Part 1

Show that

$$\int_0^{2\pi} \frac{\sin^2\theta}{2+\cos\theta} d\theta = \frac{90-52\sqrt{3}}{12-7\sqrt{3}}\pi = -2\sqrt{3}\pi + 4\pi$$

Apply the trig substitution approach:

$$\int_0^{2\pi} \frac{\sin^2\theta}{2+\cos\theta} d\theta = \oint_C \frac{\left(\frac{1}{2i}(z-\frac{1}{z})\right)^2}{2+\frac{1}{2}\left(z+\frac{1}{z}\right)} \frac{1}{iz} dz,$$

Introduce function $f(z)$

$$f(z) = \frac{\left(\frac{1}{2i}(z-\frac{1}{z})\right)^2}{2+\frac{1}{2}\left(z+\frac{1}{z}\right)} \frac{1}{iz} = \frac{-(z^2-1)^2}{z^2(z+2-\sqrt{3})\left(z+2+\sqrt{3}\right)} \frac{1}{2i}$$

Example 1. Part 2.

Consider residues:

$\mathrm{Res}(f, -2+\sqrt{3}) = \lim_{z\to -2+\sqrt{3}} \frac{z^4-2z^2+1}{z^2\left(z+2+\sqrt{3}\right)}\frac{i}{2} = i\frac{7\sqrt{3}-12}{\left(-2+\sqrt{3}\right)^2}$

$\mathrm{Res}(f, -2-\sqrt{3}) = \lim_{z\to -2-\sqrt{3}} \frac{(z^2-1)^2}{z^2\left(z+2-\sqrt{3}\right)}\frac{i}{2} = i\frac{-7\sqrt{3}-12}{\left(2+\sqrt{3}\right)^2}$ (we will not use this since $|z| \leq 1$)

$\mathrm{Res}(f, 0) = \lim_{z\to 0} \frac{d}{dz}\left(\frac{(z^2-1)^2}{(z+2-\sqrt{3})\left(z+2+\sqrt{3}\right)}\right)\frac{i}{2} = -2\frac{i}{\left(-2+\sqrt{3}\right)^2\left(2+\sqrt{3}\right)^2} = -2i$

Finally, we have

$$\int_0^{2\pi} \frac{\sin^2\theta}{2+\cos\theta}d\theta = 2\pi i\left(i\frac{7\sqrt{3}-12}{7-4\sqrt{3}} - 2i\right) = 2\pi\frac{15\sqrt{3}-26}{-7+4\sqrt{3}} = \frac{90-52\sqrt{3}}{12-7\sqrt{3}}\pi$$

Example 2. Part 1.

Show that

$$\int_0^{2\pi} \frac{\cos 3\theta}{5 - 4\cos\theta} d\theta = \frac{\pi}{12}.$$

$$\int_0^{2\pi} \frac{\cos 3\theta}{5 - 4\cos\theta} d\theta = \oint_C \frac{-1}{2z^3} \frac{z^6 + 1}{(2z-1)(z-2)} \frac{1}{i} dz$$

Consider $f(z)$

$$f = \frac{-1}{2iz^3} \frac{z^6 + 1}{(2z-1)(z-2)},$$

Example 2. Part 2.

Compute residues:

$\text{Res}(f,0) = \lim_{z\to 0} \frac{1}{2!}\frac{d^2}{dz^2}\left(\frac{-1}{2i}\frac{z^6+1}{(2z-1)(z-2)}\right) = \frac{21}{16}i$

$\text{Res}(f,\frac{1}{2}) = \lim_{z\to \frac{1}{2}} \frac{-1}{2iz^3}\frac{z^6+1}{2(z-2)} = -\frac{65}{48}i$

$\text{Res}(f,2) = \lim_{z\to 2} \frac{-1}{2iz^3}\frac{z^6+1}{(2z-1)} = \frac{65}{48}i$ (but since $|z| < 1$,we will not use this value)

Finally we can compute the integral,

$$\int_0^{2\pi} \frac{\cos 3\theta}{5-4\cos\theta}d\theta = 2\pi i\left(\frac{21}{16}i - \frac{65}{48}i\right) = \frac{1}{12}\pi$$

Example 3. Part 1.

Show that

$$\int_{-\infty}^{\infty} \frac{1}{x^6+64} dx = \frac{\pi}{48}$$

First we solve:

$$x^6 + 64 = 0$$

The solution is :

$$\{x = 2i\}, \{x = -2i\}, \left\{x = \sqrt{\left(2 - 2i\sqrt{3}\right)}\right\}, \left\{x = -\sqrt{\left(2 - 2i\sqrt{3}\right)}\right\},$$

$$\left\{x = \sqrt{\left(2 + 2i\sqrt{3}\right)}\right\}, \left\{x = -\sqrt{\left(2 + 2i\sqrt{3}\right)}\right\}$$

The function $f(z)$ is

$$f(z) = \frac{1}{x^6+64} = \frac{1}{x-2i}\frac{1}{x+2i}\frac{1}{x-\sqrt{(2-2i\sqrt{3})}}x$$

$$x\frac{1}{x+\sqrt{(2-2i\sqrt{3})}}\frac{1}{x-\sqrt{(2+2i\sqrt{3})}}\frac{1}{x+\sqrt{(2+2i\sqrt{3})}}$$

Example 3. Part 2.

Compute residues:

$\mathrm{Res}(f, 2i) = \lim_{x\to 2i} \frac{1}{x+2i} \frac{1}{x-\sqrt{(2-2i\sqrt{3})}} \frac{1}{x+\sqrt{(2-2i\sqrt{3})}} \frac{1}{x-\sqrt{(2+2i\sqrt{3})}} \frac{1}{x+\sqrt{(2+2i\sqrt{3})}} = -\frac{1}{192} i$

$\mathrm{Res}(f, -2i) = \lim_{x\to -2i} \frac{1}{x-2i} \frac{1}{x-\sqrt{(2-2i\sqrt{3})}} \frac{1}{x+\sqrt{(2-2i\sqrt{3})}} \frac{1}{x-\sqrt{(2+2i\sqrt{3})}} \frac{1}{x+\sqrt{(2+2i\sqrt{3})}} = \frac{1}{192} i$

$\mathrm{Res}\left(f, \sqrt{(2-2i\sqrt{3})}\right) = \lim_{x\to \sqrt{(2-2i\sqrt{3})}} \frac{1}{x-2i} \frac{1}{x+2i} \frac{1}{x+\sqrt{(2-2i\sqrt{3})}} \frac{1}{x-\sqrt{(2+2i\sqrt{3})}} \frac{1}{x+\sqrt{(2+2i\sqrt{3})}} = -\frac{1}{384}\sqrt{3} + \frac{1}{384} i$

$\mathrm{Res}(f, -\sqrt{(2-2i\sqrt{3})}) = \lim_{x\to -\sqrt{(2-2i\sqrt{3})}} \frac{1}{x-2i} \frac{1}{x+2i} \frac{1}{x-\sqrt{(2-2i\sqrt{3})}} \frac{1}{x-\sqrt{(2+2i\sqrt{3})}} \frac{1}{x+\sqrt{(2+2i\sqrt{3})}} = \frac{1}{384}\sqrt{3} - \frac{1}{384} i$

Example 3. Part 3.

$\mathrm{Res}\left(f, \sqrt{(2+2i\sqrt{3})}\right) =$

$\lim_{x\to\sqrt{(2+2i\sqrt{3})}} \frac{1}{x-2i}\frac{1}{x+2i}\frac{1}{x-\sqrt{(2-2i\sqrt{3})}}\frac{1}{x+\sqrt{(2-2i\sqrt{3})}}\frac{1}{x+\sqrt{(2+2i\sqrt{3})}} =$

$-\frac{1}{384}\sqrt{3} - \frac{1}{384}i$

$\mathrm{Res}\left(f, -\sqrt{(2+2i\sqrt{3})}\right) =$

$\lim_{x\to-\sqrt{(2+2i\sqrt{3})}} \frac{1}{x-2i}\frac{1}{x+2i}\frac{1}{x-\sqrt{(2-2i\sqrt{3})}}\frac{1}{x+\sqrt{(2-2i\sqrt{3})}}\frac{1}{x-\sqrt{(2+2i\sqrt{3})}} =$

$\frac{1}{384}\sqrt{3} + \frac{1}{384}i$

We only need the residues of those points in the upper half plane, and these points are:

$2i, -\sqrt{(2-2i\sqrt{3})}, \sqrt{(2+2i\sqrt{3})}$

Thus,

$$\int_{-\infty}^{\infty} \frac{1}{x^6+64}dx = 2\pi i\left(-\frac{1}{192}i + \frac{1}{384}\sqrt{3} - \frac{1}{384}i - \frac{1}{384}\sqrt{3} - \frac{1}{384}i\right) = \frac{1}{48}\pi$$

Example 4. Part 1.

Show that

$$\int_{-\infty}^{\infty} \frac{\cos(cx)}{(x^2+\alpha^2)(x^2+\beta^2)} dx = \frac{\pi}{\beta^2-\alpha^2}\left(\frac{e^{-c\alpha}}{\alpha} - \frac{e^{-c\beta}}{\beta}\right),$$

where c, α, β are positive real constants and $\alpha \neq \beta$.

What can you say about

$$\int_{-\infty}^{\infty} \frac{\sin(cx)}{(x^2+\alpha^2)(x^2+\beta^2)} dx \ ?$$

Consider

$$\int_{-\infty}^{\infty} \frac{\cos(cx)}{(x^2+\alpha^2)(x^2+\beta^2)} dx = Re\left(\int_{-\infty}^{\infty} \frac{e^{icx}}{(x^2+\alpha^2)(x^2+\beta^2)} dx\right)$$

Let f(z) be

$$f(z) = \frac{e^{icx}}{(x^2+\alpha^2)(x^2+\beta^2)},$$

Example 4. Part 1.

Computer residues:

$$\mathrm{Res}(f,\alpha i) = \lim_{x\to\alpha i} \frac{e^{icx}}{(x+\alpha i)\left(x^2+\beta^2\right)} = -\tfrac{1}{2}\tfrac{i}{\alpha(-\alpha^2+\beta^2)}e^{-\alpha c}$$

$$\mathrm{Res}(f,\beta i) = \lim_{x\to\beta i} \frac{e^{icx}}{(x+\beta i)\left(x^2+\alpha^2\right)} = \tfrac{1}{2}\tfrac{i}{\beta(-\alpha^2+\beta^2)}e^{-\beta c}$$

Then

$$\int_{-\infty}^{\infty} \frac{e^{icx}}{\left(x^2+\alpha^2\right)\left(x^2+\beta^2\right)}dx = 2\pi i\left(\frac{1}{2}\frac{i}{\beta\left(-\alpha^2+\beta^2\right)}e^{-\beta c} - \frac{1}{2}\frac{i}{\alpha\left(-\alpha^2+\beta^2\right)}e^{-\alpha c}\right) =$$

$$= \pi\frac{-e^{-\beta c}\alpha + e^{-\alpha c}\beta}{\beta\left(-\alpha^2+\beta^2\right)\alpha} = \frac{\pi}{\beta^2-\alpha^2}\left(\frac{e^{-c\alpha}}{\alpha} - \frac{e^{-c\beta}}{\beta}\right)$$

Since

$$Im\int_{-\infty}^{\infty} \frac{e^{icx}}{\left(x^2+\alpha^2\right)\left(x^2+\beta^2\right)}dx = 0,$$

$$\int_{-\infty}^{\infty} \frac{\sin(cx)}{\left(x^2+\alpha^2\right)\left(x^2+\beta^2\right)}dx = 0$$

Thank you